Release Notes xxxx-xx-xx : : not ready at all
The AD literature seems to go from miracle to failed miracle almost daily with little attempt to integrate new results with old observations. There are now a lot of data poitns to put together with cause and effect like any TV detective would do or as described in fables like the blind men and the elephant even though apparently Natalie Merchant set it to music. This work tries to fix that and finds that all these things are logically explained by well known issues such as age related digestive defects. Suggestions are further supported by in-house work on optimization of dog diets demonstrating apparent safety and utility of out of favor vitamins that could be impacted by changes in stomach chemistry. Another problem may be politics. One issue may be related to tyrosine metabolism and increased relative abundance of analogs capable of becoming toxic. This immediately jumps out as a contributor to race based AD differences not just social aspects that are touted as they are politically correct. To be sure, constituitive tyrosine metabolizers may have adaptive responses which support this flux with no net "bad" effects elsewhere and theere may turn out not to be any anwyay. In any case, its an important issue and like any other law of nature will not yield to human desire and the only way for people to benefit is follow the data. If you want to fool society, appease angry people, instead of solving problems and eliminating run on sentences for everyone turn back now to your safe space...
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Alzheimer's Disease : 10k maniacs agree its an elephant

Mike Marchywka*
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(Dated: February 4, 2026)

Effective treatments for Alzheimer's Disease(AD) have been difficult to create. Much effort has been directed at amyloid beta removal with minimal clinical benefits demonstrated. Several alternative approaches have become popular including the infection hypothesis. Since so many pathogens appear to causes AD an age related vulnerability may be a better focus. Over various time spans, a series of headlines has emerged claiming benefits for various nutrients or supplements such as thiamine, choline, lithium, or NAD+. However, none appear to generate robust recovery. Taking these results and other observations together, similar to the fable of the blind men grasping different part of an elephant, a plausible unifying theme is nutrient deficiency due to age related digestive defects. Isolated profound nutrient deficiencies will not likely exist so obvious attribution of disease to single molecular entity is not a good starting point. Note that this goes beyond failure to absorb nutrients but also the creation of age related toxins, possbly methanol for example, that may need to be mitigated. Adaptive and malapative responses may make such an initial cause difficult to discern. This work motivates "meal engineering" as an intervention strategy that may be able to correct many of these problems. If this analysis accurately captures cause and effect in the real disease, correction or prevention may be fairly easy with well known small nutrient molecules or addition of acid or chloride to meals. Individual responses to nutrient details will appear to make each case different but still possibly treatable with similar simple remedies. Indeed, there has been some limited success with dietary interventions but these fail to be designed around likely age related causes although they may coincidentally mitigate some of these problems. It may not be possible get all required nutrients from food once damage to digestive system has occured and supplements or other entities may be required. Cations such as metals, protein bound nutrients, amino acids, and lipophilic nutrients would likely be the first suspects. In particular, its likely that vitamin K and copper, along with more accepted components such as amino acids and SMVT substrates all need to be included to replace the younger absorption profile. However, brute force supplementation without regard to formulation may not work due to chemical and metabolic interactions. Two important amino acids with solubility issues in particular, tyrosine and tryptophan, and critical but reactive metals such as copper may require special attention. There are also possible political concerns if there is a hesitancy to link tyrosine metabolism to dementia. However, as always, social influences will not change the part to a best solution.

CONTENTS

I. Introduction 4

II. Common Age-Related Digestion Defects 7

III. Meal Engineering Considerations 7

IV. Candidate Ingredients 7

V. Development Paths 8

VI. Biggest Problems 8

VII. Conclusions 8

VIII. Supplemental Information 8
VIII.1. Computer Code 8

IX. Bibliography 9

References 9

Acknowledgments 13

A. Statement of Conflicts 14

B. About the Authors and Facility 14

C. Symbols, Abbreviations and Colloquialisms 14

D. General caveats and disclaimer 14

E. Citing this as a tech report or white paper 14

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@techreport{marchywka-MJM-2026-001,
filename ={elephant} ,
run-date ={February 4, 2026} ,
title ={Alzheimer's Disease : 10k maniacs agree its an elephant} ,
author ={Mike J Marchywka } ,
type ={techreport} ,
name ={marchywka-MJM-2026-001} ,
number ={MJM-2026-001} ,
version ={0.00} ,
sourcecode ={https://github.com/mmarchywka/elephant} ,
institution ={not institutionalized, independent } ,
address ={ 157 Zachary Dr Talking Rock GA 30175 USA} ,
date ={February 4, 2026} ,
startdate ={2026-01-17} ,
day ={4} ,
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pages ={ 16}
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I. INTRODUCTION

Alzheimer's disease remains an unresolved problem despite all the money and effort directed at it for the last several decades. Despite decades of development, anti-amyloid products continue to show questionable net clinical benefiti [59] and do not always get FDA approval even with creative trial design [24] . This is in spite of the fact that as early as 2002 some had noted too many inconsistent observations of amyloid or tau to be serious targets [57] .

Others are using AI for precision medicine thinking that patient heterogeneity is a problem [75].

Thinking outloud
generally with such a narrow window there is either an important constraint or it is just chasing noise.

In response, some are proposing the use of emerging AI [26] which may yield results but often is limited by traning not on raw data but human commentary and a statistical rather than "model of reality" based view.

Some are searching for connections between apparently unrelated observations [32].

Headlines sometimes appear about new scientific or anecdotal evidence pointing to a new treatment but rarely do these translate into useful therapies. The thesis of this work is that many of these results are useful but tend to be interpretted in isolation and without regard for cause and effect in real disease. "Animal models" may mimic some features of real disease but are predicated on a cause unlikely to be relevant to the pathology being treated. By coincidence, sometimes however results may be transferable but don't seem to add to understanding.

In the search for alternatives to the amyloid hypotheses, a number of other ideas have been generated. A variety of metabolic ideas have been considered.

Thinking outloud
find the work on secretase and hair color for example

Genetics have been widely explored with several common mutations linked to either familial AD(PSEN1,PSEN2, APP) or sporadic late onset (APOE) disease but with many others possibly involved [4] . Among these are things like SLC10A2 and a variety of lipid, calcium, or immune related genes. APOE is associated with many vitamins including vitamin K. APOE was identified as important for vitamin K transport to bones as early as 1996 with apoe4 noted for association with lower serum K and increased fracture risk [43] and it was noted response to supplementation occurs over months. However, a later work found "superior vitamin K status is associated with the apoE4 genotype in healthy older individuals from China and the UK" [78] .

A some human mutations identified in specific populations, along with humanized genes, have been used to create animal models of AD. The 5xFAD mouse overexpresses mutations in APP and MSEN1 and mimics some features of real disease although not NFT's [58]. EFAD mice include a human APOE gene [70] but still interpretation of any intervention has to be tempered with an understanding of the assumptions inherent in these models. Interestingly, the 5xFAD mice respond differently to diet than wild type. One study found some apparent decoupling of results,

> Thus, 5XFAD mice exhibited greater susceptibility to highfat diet than wildtype mice regarding cerebrovascular injury and cognitive impairment. On the other hand, 5XFAD mice fed highfat diet exhibited much less increase in body weight, white adipose tissue weight, and adipocyte size than their wildtype counterparts. Highfat diet significantly impaired glucose tolerance in wildtype mice but not in 5XFAD mice. Thus, 5XFAD mice had much less susceptibility to highfatdietinduced metabolic disorders than wildtype mice. [46].

These models were largely designed with the assumption that accumulation of amyloid beta was the central feature to replicate but that may turn out to not be correct. Since the cause and effect in real disease would only be captured by luck there could be a lot of limitations from any of these models.

Besdies mutation data, transcriptome analysis has found genes which are over or under transcribed in Alzheimer's Disease. As with most data classes, this has not created an unambiguous path to an intervention and cause and effect, good or bad, remains unresolved for most of these genes and their products. A gene may be over or undertranscribed depending on how it is regulated. Functional feedback could signal for more of a dysfunctional product to be created or one that can not function for some system related reason. So, typical interpretations may be missing important bpoints that may fit well into the model proposed herein.

A 2025 article found ADAMTS2 to be one of the genes robustly increased in black and European Alzheimer's brains [71]. Essentially this translates into pervasive "weak structural proteins" as described below. The same work also found in this population that GLYATL1P1 was more expressed in black AD patients than controls. AI came up with a link to starvation but I can't quite dig it out.

Another approach gaining interest is the infectious hypothesis of AD [11] [54] [12] . In support of this , a wide range of vaccines has been found to reduce risk of AD or other dementia [48]. Another study found benefits of vaccination and also anti viral treatment for herpes zoster [52]. There is also some indication AD may be transmissible similar to prion diseases [68].

However, many pathogens cause known GI damage that is often ignored or they trigger an interferon response leading to the loss of nutrients like tryptophan or nutritional immunity that may redistribute resources. Hypermetabolism has been described in Covid 19. While amyloid may have prion like properties, this kind of transmission does not seem likely in the vast majority of cases.

As no particularly good suspect has emerged, it may be worth lookin at all the data and trying to guess at age related vulnerabilities that cause more infections to lead to dementia and classic symptoms such as amyloid build ups. Nutritional defects make some sense as frailty, more than age per se, may be a correlate of age related diseases.

Works cataloging the utility of vitamins or derivatives for AD exist [72] but generally treat these things as ”one at a time” approaches. Some mixes such as a mitrochondria nutrient mix [47] have been considered.

In a prior work [50] I interpreted a brain microbiome result to describe some host factors that would explain differences in the most and least abundant organisms in AD and healthy brains post mortem. While the existence of a brain microbiome is controversial due to the expected low absolute abudance and opportunity for contamination, the analysis is somewhat robust in that contamination from other parts of the patient should still reflect differential body nutrient levels. One argument against contamination was similarity to an unrelated work on uterine microbiome and possible synergistic role of these organisms but still a lot of coincidences can occur. This work generally pointed to deficiencies in lipophilic amino acids but also suggested additional methanol in the AD patients. One likely source for this is the GI tract due to changes in microbiome or initially changes in chemistry. Other toxins could associate with methanol production too and a good strategy may be to restore GI chemistry back to youthful state.

The GLP-1 data may also motivate an interest. GLP-1’s apparently are causing non-specific nutrient deficiencies maybe similar to problems common in old age. Clinical trials showed either no benefit in AD or a positive trial [20] appeared to be cherry picked. Meanwhile anecdotes about an aged appearance (” Ozempic face”) and brain fog persist. There are a lot of lessons here but the immediate thing of relevance is that nutrient limitation that is can be ”plausibly denied” appears to generate some symptoms of aging and more to the point frailty. Interestingly, at least one study found a correlation between perceived face age and dementia risk [77].

Evidence pointing towards a cognitive benefit for GLP-1’s may be a bit misleading. In one case, they were run against sulfonylureas which have known relevant side effects. In other cases, words suggesting a cognitive benefit in actuality mean that some risk factor has been reduced without clear evidence of real clinical benefit against dementia.

Women and blacks appear to be at higher risk of AD than European males. Factors unrelated to the disease per se may confound results but that is the case with even politically correct association studies. A small 1996 study found that blacks has a higher found higher pH and HCO3 secretion as well as higher prevelance of Helicobacter pylori infection and gastritis [17]. Difference in ”normal” protein digestion in healthy males and females appear to be known and have been considered already for a basis of ”engineering foods for women” [45]. Note that this empirical difference does not rely on a specific reason although the chemistry may be informative for meal design. A small 1982 study of relatives found that older and female probands had lower acid output although some results depend on how the output is normalized [41]. One study comparing healthy 23-42 year olds with 44-71 found more acid secretion in the older group [28].

One obvious issue with skin pigment is tryosine metabolism. Both the depltion of tyrosine and generation of toxic analogs could be considered. It is possible that constituitive skin pigmentation (CSP) evolved with adaptive mechanisms that may be instructive to investigate. CSP evolution is usually described as an issue of sun exposure with damage and vitamin D synthesis being driving forces in tone [36].

While CSP may predispose to various maladies, it may also be easily treated if existing adpatations are not already active. Indeed, speculation on beneficial tyrosine analogs can not be dismissed.

Thinking outloud
goog ai went wild with ”dementia skin pigmenta

A link between dementia and melanin has been considered by many authors. A study of several thousand Taiwanese demonstrated a hazard ratio of over 12 for AD in vitiligo sufferers versus control (HR of 5.3 for any dementia) [13]. Vitiligo is an autoimmune disease making interpretation difficult but it does suggest an age related problem with melanin due to covert pathogen or defective products.

Fringe ideas in the mainstream litearture include ability of melanin to transfer light energy to the cell for ATP creation and loass impairs function [7] . More plausibly, loss of something like tyrosine or copper limits pigment synthesis in some idiosyncratic manner or more to the point consituitive pigmentation requires resources that are no longer plentiful.

In zebrafish, there was an unexpected link between PSEN2 and pigmentation [30] of unknown significance.

Tyrosine anecdotes around menopause may be one indicator. While not considered essential as it can be derived from phenyalanine, it is possible that overall supply of the precursor and regulatory systems reduce tyrosine production pathologically. This is most likely during times of ”stress” and shortages. Interestingly in connection with GLP-1 there is a tyrosine dipeptide apparently acting as a signalling molecule Amino acids such as tyrosine may have limited availability in cell cultures and dipeptides have been explored [38] for many years now. Also of interest is the need to supplement tyrosine as a dipeptide to Chinese hamseter ovary cells in bioreactors [74] with work showing possible pickiness of cells for uptake although translation to human GI tract remains to be done. Similar problems have been noted for tyrosine usage in humans from either IV injection or parenteral feeding [49]. Tyrosine-tyrosine peptide is considered a hormone and may be responsible for anorexia in urea cycle disorder patients [56].

Comparison to other racial groups does not appear informative as data are all complicated.

Some context for phenylanine to tyrosine conversion as part of the larger protein turnover dynamics has been explored [27].

Indeed GLP-1 is often considered with PYY for understanding obesity signalling [18]. While its unclear if PYY can replace the GLP-1 drugs, if supplementation is attempted it will benefit a lot from recognizing the issues explored here. Dipeptides or reproduction of "best' GI chemistry may be more important than simply adding tyrosine to diet.

Copper has a lot of motivation behind it but its rich chemistry may make it one of the more difficult to study. Copper is difficult to characterize but some results may motivate its importance. Overall it appears that AD brains are copper deficient and well formulated supplementation may be beneficial but it can not be excluded yet that other nutrients are limiting its distribution. At least one case report exists of a defect in ceruloplasmin resulting in chorea and dementia with iron deposition that resolved somewhat with fresh frozen plasma [80].

Sometimes there is a tendency to think of local levels as a passive result of supply but often complicated signalling generates overall results. As early as 2009, the paradox of excess copper with amyloid beta coincident with tissue copper deficiency was recognized and explained to some degree [34]. In prior work with dogs, the inability of chelation to produce clinical benefit in animals with hepatic copper accumulation pointed to a bottleneck in transport rather than a global excess[51]. Also in the above work it was noted that "copper toxicity" varied widely with the background diet suggesting it has to do with chemical reactions in the meal. Rats raised on a copper deficient diet showed increases in adrenal dopamine-beta-monooxygenase mRna, protein levels, and enzyme activity in decreasing order [61]. This suggests that feedback mechanisms are attempting to increase activity levels with greater transcription rates but unable to metallize the extra translation products. A 2017 mass spectrometry study on post mortem brains found overall decreased copper in AD brains similar to that in Menkes' Disease [76].

For whatever reason, vitamin K is often ignored even though interactions with APOE and appearance in "healthy" foods motivates at least an association with health.

Choline has also made recent headlines

adding to a long history for it [73] [14] and related pathways [25] including cholinesterase inhibitors [8].

Biotin and the SMVT substrates [67] have also come up in AD work and dementia more generally [69] [15] . Biotin is critical for many processes and can be obtained from digestion or intestinal microbe production [39].

Thiamine is also being considered [22] and that too is through to have acid sensitive absorption [44]. A 2012 study did show some benefits from supplementation in the elderly but due to poor absorption parental dosing was mentioned [63]. It seems that many studies on isolated nurients have come to similar conclusions but not attempted to think through cause and effect more generally. A 1988 study claimed to use a niacinamide placebo [9] which motivates another problem of an unintentionally active placebo. And in fact niacinamide has been discussed as a treatment too [42].

Lithium has recently been rediscovered [5] although positive review articles date back to at least 2012 [23] or so [53] and its not clear what happened over the intervening decade.

In 1971, it was observed that lithium could impact carbohydrate generally and inositol status in the brain [3]. In another manuscript considering inositol as a dog supplement (unpublished result)

but as of 2005 it could not be determined if lithium or inositol was more directly causal to brain changes that matter in the clinic [31]. This is likely a recurring issue in Alzheimer's and medicine overall. Associations are difficult to map to an actionable cause and effect sequence that can be beneficially modified. It is the main theme of this work that the primary cause is well removed from these associations and the elephant is not apparent from the sum of its parts. It is likely that both inositl and lithium are impacted by some common cause and simply supplementing one or the other or both can make some improvements but is a deadend to a cure.

An early proposal for vitamin K deficiency was described in 2000 citing "regulation of sulfotransferase activity and the activity of a growth factor/tyrosine kinase receptor (Gas 6/Axl) " [2].

In the last few years many works have explored vitamin K in old age conditions [21] [37] including association between vitamin K intake and cognition [10]. As bleeding or microbleeding is an issue in AD, its worth noting that 82 percent of "chronic stroke subjects" were vitamin K intake deficient and excessive vitamin E may also be a common problem [29]. The above work also describes more direct effects of vitamin K in AD including interactions with amyloid beta. (there are works that concentrate on emerging problems with vitamin E supplementation too, for example [40])

Thinking outloud

II. COMMON AGE-RELATED DIGESTION DEFECTS

To begin to make sense of the above and look towards the design of a better intervention, some knowledge of likely defects and resulting deficiencies needs to be established.

By 1992, common themes in impairment reports were emerging. One review suggested decreased taste and hypochlorhydria with decreased uptake of B-12 and minerals but increased absorption of lipids [65]. A small 1997 study of independently living people over 65 found about 90 percent had basal pH under 3.5 [35] although there is no indication of how this compares to the young or very old.

A study of 79 healthy elderly people found some indications of lower acidity than young people but did not consider it relevant in most individuals at least for drug absorption [66].

The pancreas and intestines may also be effected. One review mentions decreases in calcium, zinc, and magnesium but not copper in old rats [33]. However, the pancreas and intestines must work with whatever the stomach produces with the meal. Earlier, I considered that lack of an early low pH step may produce more idiosyncratic results as some mineral can irreversibly precipitate [51].

Around 2004, it was thought that reserve capacity was sufficient although some reduced absorption of vitamins such as A,D,K, and B6 but tended to minimize significance for treating aging [19]. One problem however is that even today sufficient intake and uptake may not be clear and the impact of multiple ”low levels” could invoke adpative responses.

Today however protein deficiency appears to be reasonably well accepted as an issue with aging and its manifestation as sarcopenia [62]. What mey be less apparent is any changes in quality such as due to translational infidelity or lack of ”other stuff.” Sarcopenia or frailty appear to be highly correlated with vulnerability to several ”age related” conditions [79].

Further, the impact of drugs common in the elderly on nutrient status is becoming recognized [6].

III. MEAL ENGINEERING CONSIDERATIONS

Based on the above, the goal then is to replace nutrients and remove toxics that are more common in old age. Return to a youthful GI tract may be a reasonable goal although it may be possible to do even better. Conceivably if nutrient uptake can overcome GI limitations some healing may be observed making possible a return to normal food for a while.

Nutrients need only be supplied over ”relevant” time scales. This can vary a lot from nutrient to nutrient and allows better segregation and rotation than trying to get everything everyday. Incompatibilities include chemistry and competition for receptors or enzymes or transporters etc.

Not every meals needs to be the same and in fact currently with the dogs one meal contains most things while a second is largely reserved for copper.

IV. CANDIDATE INGREDIENTS

Candidates are drawn from prior thoughts on food and vitamin supplements along with the presumed GI disturbances common in old age.

Any foods or supplements reputed to improve some age related condition could be examined. In many cases the folklore appears to be a useful starting point as it may rationalize well with the above theory or be supported by in-house work here with dog feeding.

Fermented foods meet many of the criteria for obtaining nutrients with age impaired digestions. They tend to have low pH, and can even include pH as part of the definition, and essential molecules such as free amino acids and vitamin k.

Olive oil has been an important part of in-house dog diets (manuscript in preparation) as well as being a highlighted component of the Mediterranean Diet. A variety of possible causal pathways could be imagined for it but here it is considered somewhat uniquely, along with lecithin and phosphorous sources, as an absorption enhancer. Its interesting to note, although of unknown significance, that yeast with a higher amount of membrane oleic acid have higher ethanol tolerance [81]. As part of a meal formulation it may help with uptake of lipophilic nutrients.

2 related objectives are to improve chloride levels and reduce pH. In the results cited previously, acid came from citric acid and HCl supplement formulations. Formulations that use HCl as a raw component would require great care and may be avoided initially. Some soft drinks such as diet cola beverages contain phosphoric acid that may have some benefits. Chloride was largely from potassium chloride and again added HCl.

Title			
Ingredient	Amount	InCompatible	Remarks

TABLE I.

V. DEVELOPMENT PATHS

Since many candidate components are GRAS, its possible to outline informal or "crowd sourced" data from motivated parties such as caregivers. Concerns about malicious data due to various incentives as well as other issues would have to be considered in detail. I have been using MUQED daily to record dog diets and relate to outcomes. This helped to identify olive oil as an improtant component although it may have a long response time if other sources of oleic acid such as chicken are included in the diet. The added workflow just requires a few minutes after each meal or dosing to enter whatever was given and have it checked for basic syntax. I use "vi" although more sophisticated or voice entry may be easy to do. The current implementation of MUQED creates a small vocabulary of foods, supplements and outcomes as well as a simply syntax for entering amounts right after or even skeletons before consumption. As the current implementation is designed for just "a few" people or dogs, everything is done with text files locally although collation with a standard vocabulary could be done with better scaling implementations.

My in-house results with dogs does suggest some folk lore or anecdotes may have useful signals but will require more data to sort out and a MUQED-like approach may fill the void.

Even if this fails to mitigate disease, it may not be a frivolous effort especially if any trends are observed with prescription medicines.

VI. BIGGEST PROBLEMS

Thinking outloud
lot of issues here include off-target effects, accurate prescribing patterns, population differences between users and non uers, any real effect etc. AGA seems to be pushing this things probably get ad revenue lol

As proton concentration is thought to be a big issue, it may be surprising that chronic PPI use does not lead to obvious epidemic of AD. Although in older people they may not have much real impact as stomach acid is already likely to be low (if PPI's are effective at anything it is likely through some other mechanism. If you can't tolerate stomach acid you are going to be malnourished anyway - something is wrong so you may expect accurately prescribed PPI users to already be predisposed to dementia). Current evidence is conflicting even among older or sick people who may already have GI impairments minimizing any effect of PPI's. A 2023 study of a 1.98 million Danish cohort found an incidence of any dementia ratio between PPI users and non-users of 1.36 but reducing with age [60]. The American Gastroenterological Association appears to advance the position that this is unlikely. One post hoc analysis of the ASPREE aspirin trial involving people older than 65, found no association between PPI use and dementia [55]. Reported results are heavily corrected for various factors and the raw HR was not immediately obvious. It is possible the low dose aspirin has some impact but difficult to tell as presented.

Protons and chloride appear to be differentially regulated and PPI's do not stop normal chloride secretion [16] despite the fact they may be used to prevent excess fecal chloride excretion in abnormal cases [1]. There are several chloride transporters in the stomach including calcium activated [64] that may decouple from PPI's. The cystic fibrosis literature may provide some clues.

However, the theory here is a bit empirical and recognizes other factors than pH alone may be important. With a non-zero supply of most required molecules adaptive responses may be able to move the effects of the deficiencies around making the clinical presentation difficult to trace to original problem.

Even if the theory is sound and eventually is proven useful as a starting point, social issues are a big problem. For whatever reason amyloid drugs appear to be getting a more favorable response than is supported by known data. Anything involving skin pigment will attract angry people.

VII. CONCLUSIONS

VIII. SUPPLEMENTAL INFORMATION

VIII.1. Computer Code

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ACKNOWLEDGMENTS

1. Pubmed eutils facilities and the basic research it provides.

2. Free software including Linux, R, LaTex etc.

3. Thanks everyone who contributed incidental support.

Appendix A: Statement of Conflicts

No specific funding was used in this effort and there are no relationships with others that could create a conflict of interest. I would like to develop these ideas further and have obvious bias towards making them appear successful. Barbara Cade, the dog owner, has worked in the pet food industry but this does not likely create a conflict. We have no interest in the makers of any of the products named in this work.

Appendix B: About the Authors and Facility

This work was performed at a dog rescue run by Barbara Cade and housed in rural Georgia. The author of this report ,Mike Marchywka, has a background in electrical engineering and has done extensive research using free online literature sources. I hope to find additional people interested in critically examining the results and verify that they can be reproduced effectively to treat other dogs.

Appendix C: Symbols, Abbreviations and Colloquialisms

TERM	definition and meaning

Appendix D: General caveats and disclaimer

This document was created in the hope it will be interesting to someone including me by providing information about some topic that may include personal experience or a literature review or description of a speculative theory or idea. There is no assurance that the content of this work will be useful for any paricular purpose.

All statements in this document were true to the best of my knowledge at the time they were made and every attempt is made to assure they are not misleading or confusing. However, information provided by others and observations that can be manipulated by unknown causes (”gaslighting”) may be misleading. Any use of this information should be preceded by validation including replication where feasible. Errors may enter into the final work at every step from conception and research to final editing.

Documents labelled ”NOTES” or ”not public” contain substantial informal or speculative content that may be terse and poorly edited or even sarcastic or profane. Documents labelled as ”public” have generally been edited to be more coherent but probably have not been reviewed or proof read.

Generally non-public documents are labelled as such to avoid confusion and embarassment and should be read with that understanding.

Appendix E: Citing this as a tech report or white paper

Note: This is mostly manually entered and not assured to be error free.
This is tech report MJM-2026-001.

Version	Date	Comments
0.01	2026-01-17	Create from empty.tex template
-	February 4, 2026	version 0.00 MJM-2026-001
1.0	20xx-xx-xx	First revision for distribution

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Version	Date	URL

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startdate ={2026-01-17} ,
day ={4} ,
month ={2} ,
year ={2026} ,
author1email ={marchywka@hotmail.com} ,
contact ={marchywka@hotmail.com} ,
author1id ={orcid.org/0000-0001-9237-455X} ,
pages ={ 16}
}

Supporting files. Note that some dates,sizes, and md5's will change as this is rebuilt.

This really needs to include the data analysis code but right now it is auto generated picking up things from prior build in many cases

```
2703 Feb 3 13:10 comment.cut f995244b5e95cab8bb975bbef4509f63
11639 Feb 3 13:10 elephant.aux 459edc399e374d53031ddfc798b20c02
35318 Feb 3 13:10 elephant.bbl 94423c4ce3a1cf85c2e973828c80dae7
351736 Feb 3 13:09 elephant.bib 371dfcfe58e42825d9a129c1c8644fe9
1606 Feb 3 13:10 elephant.blg f5be3c060855cbda3a816f0c4a0dcc50
0 Feb 3 13:10 elephant.bundle_checksums d41d8cd98f00b204e9800998ecf8427e
25284 Feb 3 13:10 elephant.fls c191be6b918cf7e5b99e8e136646d6ad
3 Feb 3 13:10 elephant.last_page 8c9eb686bf3eb5bd83d9373eadf6504b
50305 Feb 3 13:10 elephant.log 2d7e0605d84f84fc230a73dac67b2053
0 Feb 3 13:10 elephantNotes.bib d41d8cd98f00b204e9800998ecf8427e
3546 Feb 3 13:10 elephant.out 12733f2f81d87bcbb721796d6792973a
171699 Feb 3 13:10 elephant.pdf 342e06c74921fd564bfa88cb01db8dfb
44555 Feb 3 13:09 elephant.tex 5329650acb38188c4575c6e481bc741a
1652 Feb 3 13:10 elephant.toc 4f246bc3892fe3572808ec9dd4c3d6e6
49863 Jan 17 03:59 /home/documents/latex/bib/mjm_tr.bib e93b93c88b4601ca44b7886f378f254a
48988 Jul 1 2025 /home/documents/latex/bib/releases.bib f2fdf87a36feddccbab0918dc2b00129
7331 Jan 24 2019 /home/documents/latex/pkg/fltpage.sty 73b3a2493ca297ef0d59d6c1b921684b
7434 Oct 21 1999 /home/documents/latex/pkg/lgrind.sty ea74beead1aa2b711ec2669ba60562c3
1069 Oct 15 2021 /home/documents/latex/share/includes/disclaimer-gaslight.tex 94142
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425 Oct 11 2020 /home/documents/latex/share/includes/disclaimer-status.tex b276f09e06a3a9114f927e4199f379f7
3927 May 2 2024 /home/documents/latex/share/includes/mjmaddbib.tex d58183f98615737b38f9d6e03fc064de
117 Oct 16 2023 /home/documents/latex/share/includes/mjmlistings.tex c1c3bd564b7fb321b4d8f223c7d3cde3
4867 Jun 23 2025 /home/documents/latex/share/includes/mycommands.tex d78deef4710745e9b4b29511fd3b610d
1031 Nov 8 07:26 /home/documents/latex/share/includes/myrevtexheaders.tex 79e40ba4aee901851cf9a06010aed8c2
2919 Jun 9 2024 /home/documents/latex/share/includes/myskeletonpackages.tex 08
    a01c33af793bfeb936c6684d2e792c
1538 Aug 14 2021 /home/documents/latex/share/includes/recent_template.tex 49763d2c29f74e4b54fa53b25c2cc439
7217 May 15 2023 /home/documents/latex/share/overrides/mol2chemfig.sty 5aa81c197d462c2153a794ccee9881ef
311056 Feb 3 13:09 non_pmc_elephant.bib ac9b341945ab0b9f89ead9638fb92cba
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40269 Feb 3 12:12 pmc_elephant.bib 405cec971734112b3c43f60e4a4af370
413 Jan 17 03:59 releases.tex 56213b8bd7e164dd6e668b62f5acf868
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33495 Jan 30 2023 /usr/share/texlive/texmf-dist/bibtex/bst/urlbst/plainurl.bst 46
    cfb45e24a877059f9f7e142c7fc61c
165211 Jul 10 2025 /var/lib/texmf/fonts/map/pdftex/updmap/pdftex.map 6b08d096f3d0fe5ff6b266061a2b45e0
5712750 Jul 10 2025 /var/lib/texmf/web2c/luahbtex/lualatex.fmt a41ebf2b4a8fce76020b8e7c277eca3a
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